ET AI HACKATHON 2026 · PROBLEM STATEMENT 4

PRAMAAN

EPC Deviation Intelligence for Data-Centre Delivery

*Pramaan (प्रमाण) — Sanskrit for **proof**. A benchmarked prototype: the deterministic numbers reproduce offline; live-model results are reported honestly, not as a fixed score.*

0.862
mean recall
BENCHMARKED V1.2

53
spec–submittal pairs
TEAM-AUTHORED

1
LLM reasoning core
+ DET. SERVICES

parth-tan.vercel.app/judge · parth-tan.vercel.app/evidence

THE PROBLEM

A Late Submittal Deviation Becomes a Commissioning & Schedule Failure

When a vendor submittal quietly fails the spec, the miss usually surfaces at **commissioning** — after fabrication, shipping, and install — as rework, a slipped ready-for-service date, and re-procurement of long-lead gear.

Caught at commissioning

Weeks 16–44

Rework, schedule slip, long-lead re-order — the most expensive place to find it

Caught at submittal review

Week 11

One RFI, on the day the document lands — while it still costs almost nothing to fix

Weeks shown are the demo project (Meghdoot) scenario. SCENARIO

WHY CURRENT REVIEW FAILS

Three Documents, Three Parties, No Shared View

Owner requirement

The design basis and governing standards — what was actually specified.

Vendor submittal

Datasheets and cut sheets — thousands of pages, buried values, prose and tables.

Commissioning & schedule

The Cx test that will fail and the milestone that slips — owned by yet another team.

A 7-minute battery where the spec needs 10 hides in the gaps between them — until an integrated systems test fails at Week 38.

THE SOLUTION

Requirement → Deviation → Commissioning Test → Action Window

Pramaan reconciles the owner requirement against the vendor submittal and the governing standards, then traces each deviation to the test it fails and the lead time to fix it.

Owner requirement + standards + **Vendor submittal** PDF · image · text → DEVIATION **7 vs 10 min** cited to the clause → CX TEST **IST-07** it will fail → WINDOW **27 weeks** to fix first

Example values from the demo pair. SCENARIO The engine detects them from the raw documents, not a lookup.

LIVE PRODUCT · JUDGE MODE

See It Reason in Under a Minute

/judge — one screen, two buttons, honest status.

Load deviation demo ★

A realistic design basis vs a vendor submittal in natural prose. Pramaan surfaces a hidden 2N→N+1 and a 10-min→8-min non-compliance, each with its standard, Cx test, and lead time.

Load compliant demo ✓

A fully compliant submittal — the correct answer is **zero deviations**. It shows Pramaan does not cry wolf, the behaviour the clean-negative controls measure.

- Token-by-token reasoning stream, then a cited deviation register.
- **Truthful provenance chip** on every result: live LLM · vision · deterministic rule floor · OCR.
- Upload your own spec + submittal (PDF / image / text) — nothing is pre-seeded.

ARCHITECTURE — THE TRUTH

One Compliance Reasoning Graph, Not Five LLM Agents

A LangGraph reasoning graph with two bounded cycles + connected deterministic services + a reliability layer. **Exactly one node reasons with an LLM.**

DET	LLM CORE	DET · CYCLE 1	DET · CYCLE 2	RULE
Ingest	**Reconcile**	**Retrieve**	**Critique**	**Cx Predict**
PDF/image → text	extract + reason	fetch standard	self-check	test + lead time

Ingest → Reconcile ↺ Retrieve ↺ Critique → Cx Predict

Reasoning graph

Retrieval tool-call + self-critique reflexion cycles, both bounded. Falls back to a sequential runner.

Deterministic services

Cx mapping · schedule (CPM + Monte Carlo) · supply-chain · project graph. RFI copilot is a service, not a graph node.

Reliability layer

Provider failover for availability → deterministic rule floor. OCR via `/ocr-check`. Rate limits + upload validation.

BENCHMARK PROOF

ps4_external_v1 (v1.2) — Frozen, Provenance-Tracked

53 spec–submittal pairs · 129 frozen labels · 17 systems · 64 clean negatives · 3-pass featured run (`gemini-3.1-flash-lite`) BENCHMARKED

0.862
mean recall (0.841–0.873)

0.953
mean precision

0.905
mean F1

0.000
false-alert rate · 64 clean negatives

~2.5s
p50 latency / pair

0.111
rule-engine baseline recall DETERMINISTIC

605
automated tests

A benchmark result — **not** real-world-accuracy or field validation. The LLM core clears the deterministic floor (0.862 vs 0.111) on the same frozen labels.

TRUST & LIMITATIONS

Every Number Carries Its Caveat

What we can stand behind

- 0 false alerts on 64 clean-negative controls
- Every finding cites clause + standard + submittal
- Frozen labels + hashed sources, reproducible offline
- Provenance shown; rule-floor 0-result is not a green pass

What we do not claim

- Team-authored fixtures — 10 primary-source-derived (5 verified URLs) STORED PDFS PENDING
- Labels single-author frozen — **reviewer-2 human adjudication pending**
- Automated cross-check (123/129) is machine QA, **not** a human reviewer
- Not field- or customer-validated; no ROI claim

Full register + live status: **/evidence** · `docs/CLAIMS_REGISTER.md`

RELIABILITY · BUILT FOR THE BAD DAY

Availability Failover, OCR, and a Deterministic Floor

Provider failover

gemini → Qwen gateway → Groq → Claude → local → rule floor. Only configured providers are tried. **Availability, not accuracy** — every leg scored the same. Live at `/llm-check`.

Deterministic floor

No silent zeros: if no model answers, the rule engine still returns headline deviations from the documents. Low-recall by design (0.111) — and labelled as such in the UI.

OCR + demo hardening

Scanned-PDF / image OCR where Tesseract is installed — ground truth at `/ocr-check`. Optional token auth, per-IP rate limits, upload validation.

DEMO HARDENING, NOT PRODUCTION

`/health` shows the running commit · every endpoint returns 200 without a key.

BUSINESS IMPACT

The Value Is Time, Caught Early — Not a Fabricated ROI

Moving detection from commissioning to submittal-day review turns a seven-figure slip into a one-line RFI. The defensible unit is **lead-time weeks**, not a customer dollar figure.

Weeks 16–44
where these deviations otherwise surface SCENARIO

Week 11
caught at submittal review

27 wk
lead time on the hero UPS deviation SCENARIO

Any cost figure in the app is an explicitly-labelled illustrative scenario under stated assumptions — **not** a measured customer ROI. Manual-hours reduction is not yet measured.

ROADMAP

From Benchmarked Prototype to Validated Product

Evidence depth

- Stored primary-source datasheet PDFs
- Expand primary-source-derived pairs
- Larger, harder benchmark slices

Independent review

- Reviewer-2 two-person adjudication
- Resolve the 6 audit-flagged labels
- Practitioner (CxA) validation

Production hardening

- Shared-store rate limiting + auth
- pgvector/Qdrant + async at scale
- Delta-only re-checks, drawings via vision

Ordered by what most increases trust first. PENDING

EVIDENCE BEFORE CONFIDENCE

PRAMAAN

Benchmark v1.2: recall 0.862 · precision 0.953 · F1 0.905 · 0 false alerts on 64 clean negatives

One reasoning graph · deterministic services · reliability layer — reviewer-2 pending, and we say so.

▶ Judge Mode

Evidence & Proof

github.com/bansalbhunesh/parth · parth-tan.vercel.app · parth-1-ma30.onrender.com